

Addition and subtraction

$2 + 3 = 5$

$7 + 4 = 11$

$9 + 5 = 14$

$11 + 7 = 18$

$5 + 8 = 13$

$15 + 4 = 19$

$7 + 10 = 17$

$7 + 2 = 9$

$3 - 2 = 1$

$15 - 4 = 11$

$9 - 7 = 2$

$5 - 2 = 3$

$20 - 10 = 10$

$18 - 8 = 10$

$10 - 9 = 1$

$13 - 7 = 6$

Example 1: Find the following:

(i) $60 + 20$

Solution:

$$\begin{array}{r} 60 \\ + 20 \\ \hline 80 \end{array}$$

or

$60 + 20 = 80$

(ii) $72 + 9$

Solution:

$$\begin{array}{r} \textcircled{1} \\ 72 \\ + 9 \\ \hline 81 \end{array}$$

or

$72 + 9 = 81$

(iii) $39 + 11$

Solution:

$$\begin{array}{r} \textcircled{1} \\ 39 \\ + 11 \\ \hline 50 \end{array}$$

or

$39 + 11 = 50$

Example 2: Find the following:

(i) $77 - 37$

Solution:

$$\begin{array}{r} 77 \\ - 37 \\ \hline 40 \end{array}$$

or

$77 - 37 = 40$

(ii) $96 - 24$

Solution:

$$\begin{array}{r} 96 \\ - 24 \\ \hline 72 \end{array}$$

or

$96 - 24 = 72$

(iii) $58 - 20$

Solution:

$$\begin{array}{r} 58 \\ - 20 \\ \hline 38 \end{array}$$

or

$58 - 20 = 38$



Activity

Complete the following addition and subtraction facts to 100.

$$50 + 32 = 82$$

$$82 - 50 = 32$$

$$32 + 50 = \square$$

$$82 - 32 = \square$$

$$69 + 11 = \square$$

$$80 - 11 = \square$$

$$80 + \square = 100$$

$$100 - \square = 80$$

$$\square + 15 = 65$$

$$65 - \square = 15$$

$$92 + 8 = \square$$

$$100 - 8 = \square$$

Exercise 1

1. Match the following.

$32 + 64$

$19 + 21$

$61 + 10$

$91 + 9$

$58 + 12$

$50 + 50$

$9 + 91$

$80 + 20$

$60 + 10$

$60 + 11$

$30 + 10$

$64 + 32$

2. Complete the following addition and subtraction facts

$72 + 8 = 80$

$63 + 7 = \square$

$90 - \square = 80$

$50 - 22 = \square$

$100 + 0 = \square$

$21 + 41 = \square$

$80 - 72 = 8$

$70 - 63 = \square$

$80 + \square = 90$

$22 + 28 = \square$

$100 - \square = 100$

$62 - 41 = \square$

Add and subtract

Mental addition of 3-digit number with 1-digit number.

Without regrouping	With regrouping
Example: Add 245 with 3 Solution: $245 + 3 = 248$	Example: Add 326 and 7 Solution: By regrouping $326 + 7 = \boxed{326 + 4} + 3$ $= 333$

Mental addition of 3-digit number with tens.

Without regrouping	With regrouping
Example: Add 534 with 20 Solution: $534 + 20 = 554$	Example: Add 937 with 40 Solution: By regrouping $937 + 40 = \boxed{930 + 40} + 7 = 977$

Mental addition of 3-digit number with 2-digit number.

Without regrouping	With regrouping
Example: Add 400 with 40 Solution: $400 + 40 = 440$	Example: Add 564 with 17 Solution: By regrouping $564 + 17 = \boxed{564 + 6} + 11$ $= 581$

Mental addition of two 3-digit numbers without regrouping

Without regrouping	With regrouping
e.g. $520 + 260$ Solution: $520 + 260 = 780$	e.g. $208 + 300$ Solution: $= 8 + 200 + 300$ $= 8 + 500$ $= 508$



Activity

Add the following mentally

$206 + 5 = \boxed{211}$

$500 + 400 = \boxed{900}$

$720 + 350 = \boxed{}$

$110 + 760 = \boxed{}$

$226 + 7 = \boxed{}$

$196 + 14 = \boxed{}$

$180 + 30 = \boxed{}$

$180 + 9 = \boxed{}$

$560 + 7 = \boxed{}$

$972 + 18 = \boxed{}$

$921 + 11 = \boxed{}$

$442 + 5 = \boxed{}$

$192 + 28 = \boxed{}$

$500 + 18 = \boxed{}$

Addition of 3-digit number with 1-digit number in written form.

Without regrouping	With regrouping																																											
Example: Add 523 with 5 Solution: <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td>5</td><td>2</td><td>3</td></tr><tr><td>+</td><td></td><td>5</td></tr><tr><td colspan="3"> </td></tr><tr><td>5</td><td>2</td><td>8</td></tr></table>	H	T	O	5	2	3	+		5				5	2	8	Example: Add 627 with 5 Solution: <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td></td><td>①</td><td></td></tr><tr><td>6</td><td>2</td><td>7</td></tr><tr><td>+</td><td></td><td>5</td></tr><tr><td colspan="3"> </td></tr><tr><td>6</td><td>3</td><td>2</td></tr></table> <table><tr><td colspan="2">Ones</td></tr><tr><td></td><td>7</td></tr><tr><td>+</td><td>5</td></tr><tr><td colspan="2"> </td></tr><tr><td>①</td><td>2</td></tr></table>	H	T	O		①		6	2	7	+		5				6	3	2	Ones			7	+	5			①	2
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5	2	3																																										
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Ones																																												
	7																																											
+	5																																											
①	2																																											
Note: We just add ones.	Steps: (i) Add ones to get 12. (ii) Carry 1 to ten's place. (iii) Add tens and get 3. Note: Regrouping refers to "carrying forward" in addition																																											

Addition of 3-digit number with tens in written form

Without regrouping	With regrouping																																											
Example: Add 523 with 60 Solution: <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td>4</td><td>2</td><td>3</td></tr><tr><td>+</td><td>6</td><td>0</td></tr><tr><td colspan="3"> </td></tr><tr><td>4</td><td>8</td><td>3</td></tr></table>	H	T	O	4	2	3	+	6	0				4	8	3	Example: Add 634 with 80 Solution: <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td>①</td><td></td><td></td></tr><tr><td>6</td><td>3</td><td>4</td></tr><tr><td>+</td><td>8</td><td>0</td></tr><tr><td colspan="3"> </td></tr><tr><td>7</td><td>1</td><td>4</td></tr></table> <table><tr><td colspan="2">Tens</td></tr><tr><td></td><td>3</td></tr><tr><td>+</td><td>8</td></tr><tr><td colspan="2"> </td></tr><tr><td>①</td><td>1</td></tr></table>	H	T	O	①			6	3	4	+	8	0				7	1	4	Tens			3	+	8			①	1
H	T	O																																										
4	2	3																																										
+	6	0																																										
4	8	3																																										
H	T	O																																										
①																																												
6	3	4																																										
+	8	0																																										
7	1	4																																										
Tens																																												
	3																																											
+	8																																											
①	1																																											
Note: We just add tens.	Steps: (i) Add tens to get 11. (ii) Carry 1 to hundred's place. (iii) Add hundreds to get 7.																																											

Addition of 3-digit number with 2-digit number in written form

Without regrouping	With regrouping
Example: Add 721 with 36 Solution: $ \begin{array}{r} \text{H T O} \\ 721 \\ + 36 \\ \hline 757 \end{array} $ Note: We just add units and tens separately.	Example: Add 567 with 58 Solution: $ \begin{array}{r} \text{H T O} \\ \textcircled{1} \textcircled{1} \\ 567 \\ + 58 \\ \hline 625 \end{array} $ Steps: (i) Add ones to get 15. (ii) Carry 1 to tens place. (iii) Add tens to get 12. (iv) Carry 1 to hundreds place. (v) Add hundreds to get 6.

Addition of two 3-digit numbers in written form

Without regrouping	With regrouping
Example: Add 543 with 316 Solution: $ \begin{array}{r} \text{H T O} \\ 543 \\ + 316 \\ \hline 859 \end{array} $ Note: Add every place value separately.	Example: Add 475 with 348 Solution: $ \begin{array}{r} \text{H T O} \\ \textcircled{1} \textcircled{1} \\ 475 \\ + 348 \\ \hline 823 \end{array} $ Steps: (i) Add ones to get 13. (ii) Carry 1 to tens place. (iii) Add tens to get 12. (iv) Carry 1 to hundreds place. and add hundreds to get 8.



Activity Add the following

$$\begin{array}{r} 125 \\ + 3 \\ \hline \end{array}$$

$$128$$

$$\begin{array}{r} 346 \\ + 30 \\ \hline \end{array}$$

$$376$$

$$\begin{array}{r} \textcircled{1} \textcircled{1} \\ 296 \\ + 25 \\ \hline \end{array}$$

$$321$$

$$\begin{array}{r} \textcircled{1} \textcircled{1} \\ 289 \\ + 346 \\ \hline \end{array}$$

$$635$$

$$\begin{array}{r} 236 \\ + 8 \\ \hline \end{array}$$

$$\boxed{}$$

$$\begin{array}{r} 529 \\ + 80 \\ \hline \end{array}$$

$$\boxed{}$$

$$\begin{array}{r} 783 \\ + 16 \\ \hline \end{array}$$

$$\boxed{}$$

$$\begin{array}{r} 537 \\ + 152 \\ \hline \end{array}$$

$$\boxed{}$$

$$\begin{array}{r} 562 \\ + 396 \\ \hline \end{array}$$

$$\boxed{}$$

$$\begin{array}{r} 918 \\ + 32 \\ \hline \end{array}$$

$$\boxed{}$$

$$\begin{array}{r} 639 \\ + 80 \\ \hline \end{array}$$

$$\boxed{}$$

$$\begin{array}{r} 696 \\ + 23 \\ \hline \end{array}$$

$$\boxed{}$$

Exercise 2

1. Add mentally.

(i) $123 + 5 =$

$$\boxed{}$$

(ii) $634 + 8 =$

$$\boxed{}$$

(iii) $579 + 20 =$

$$\boxed{}$$

(iv) $440 + 80 =$

$$\boxed{}$$

(v) $810 + 23 =$

$$\boxed{}$$

(vi) $913 + 29 =$

$$\boxed{}$$

(vii) $638 + 151 =$

$$\boxed{}$$

(viii) $785 + 136 =$

$$\boxed{}$$

(ix) $513 + 64 =$

$$\boxed{}$$

(x) $345 + 9 =$

$$\boxed{}$$

2. Add the following.

(i) $\begin{array}{r} 236 \\ + 53 \\ \hline \end{array}$

(ii) $\begin{array}{r} 986 \\ + 3 \\ \hline \end{array}$

(iii) $\begin{array}{r} 508 \\ + 6 \\ \hline \end{array}$

(iv) $\begin{array}{r} 518 \\ + 43 \\ \hline \end{array}$

(v) $\begin{array}{r} 675 \\ + 20 \\ \hline \end{array}$

(vi) $\begin{array}{r} 853 \\ + 70 \\ \hline \end{array}$

(vii) $\begin{array}{r} 834 \\ + 76 \\ \hline \end{array}$

(viii) $\begin{array}{r} 456 \\ + 168 \\ \hline \end{array}$

(ix) $\begin{array}{r} 345 \\ + 632 \\ \hline \end{array}$

(x) $\begin{array}{r} 278 \\ + 369 \\ \hline \end{array}$

(xi) $\begin{array}{r} 509 \\ + 198 \\ \hline \end{array}$

(xii) $\begin{array}{r} 467 \\ + 298 \\ \hline \end{array}$

Mental subtraction of 1-digit number from 3-digit number.

Without regrouping	With regrouping
Example: Subtract 4 from 325 Solution: $325 - 4 = 321$ Note: We just subtract ones	Example: Subtract 6 from 534 Solution: By regrouping $534 - 6 = 524 + \boxed{10 - 6}$ $= 528$

Mental subtraction of tens from 3-digit number.

Without regrouping	With regrouping
Example: Subtract 50 from 678 Solution: $678 - 50 = 628$ Note: We only subtract tens	Example: Subtract 70 from 624 Solution: By regrouping $624 - 70 = 524 + \boxed{100 - 70}$ $= 524 + 30$ $= 554$

Mental subtraction of 2-digit number from 3-digit number.

Without regrouping	With regrouping
Example: Subtract 35 from 978 Solution: $978 - 35 = 943$ Note: We only subtract ones and tens separately	Example: Subtract 65 from 542 Solution: By regrouping $542 - 65 = 442 + \boxed{100 - 65}$ $= 442 + 35$ $= 477$

Mental subtraction of two 3-digit numbers.

Without regrouping	With regrouping
Example: Subtract 567 from 879 mentally. Solution: $879 - 567 = 312$ Note: We only subtract digits of respective place values in mind.	Example: Subtract 255 from 620 mentally. Solution: By regrouping in mind $620 - 255 = \boxed{520 - 200} + \boxed{100 - 55}$ $= 320 + 45$ $= 365$



Activity Subtract mentally.

$527 - 3 = \boxed{524}$

$478 - 50 = \boxed{428}$

$690 - 25 = \boxed{665}$

$500 - 130 = \boxed{370}$

$916 - 9 = \boxed{}$

$515 - 40 = \boxed{}$

$735 - 14 = \boxed{}$

$790 - 280 = \boxed{}$

$796 - 87 = \boxed{}$

$340 - 165 = \boxed{}$

$506 - 30 = \boxed{}$

$800 - 210 = \boxed{}$

Subtraction of 1-digit number from 3-digit number in written form.

Without regrouping	With regrouping																																					
Example: Subtract 5 from 398 Solution: <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td>3</td><td>9</td><td>8</td></tr><tr><td>-</td><td></td><td>5</td></tr><tr><td colspan="3"> </td></tr><tr><td>3</td><td>9</td><td>3</td></tr></table>	H	T	O	3	9	8	-		5				3	9	3	Example: Subtract 8 from 275 Solution: <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td></td><td>⑥</td><td>⑮</td></tr><tr><td>2</td><td>7</td><td>5</td></tr><tr><td></td><td>-</td><td>8</td></tr><tr><td colspan="3"> </td></tr><tr><td>2</td><td>6</td><td>7</td></tr></table> <table><tr><td>Ones</td></tr><tr><td>1 5</td></tr><tr><td>- 8</td></tr><tr><td>7</td></tr></table>	H	T	O		⑥	⑮	2	7	5		-	8				2	6	7	Ones	1 5	- 8	7
H	T	O																																				
3	9	8																																				
-		5																																				
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Ones																																						
1 5																																						
- 8																																						
7																																						
Note: We only subtract ones.	Steps: (i) We can not subtract 8 from 5. We borrow 1 ten leaving behind 6 tens (ii) Now, we subtract 8 from 15 to get 7 and write the remaining digits same. Note: Regrouping refers to "Borrowing" in subtraction.																																					

Subtraction of tens from 3-digit number in written form.

Without regrouping	With regrouping																																						
Example: Subtract 20 from 694 Solution: <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td>6</td><td>9</td><td>4</td></tr><tr><td>-</td><td>2</td><td>0</td></tr><tr><td colspan="3"> </td></tr><tr><td>6</td><td>7</td><td>4</td></tr></table>	H	T	O	6	9	4	-	2	0				6	7	4	Example: Subtract 70 from 526 Solution: <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td>④</td><td>⑫</td><td></td></tr><tr><td>5</td><td>2</td><td>6</td></tr><tr><td>-</td><td>7</td><td>0</td></tr><tr><td colspan="3"> </td></tr><tr><td>4</td><td>5</td><td>6</td></tr></table> <table><tr><td>Tens</td></tr><tr><td>1 2</td></tr><tr><td>- 7</td></tr><tr><td> </td></tr><tr><td>5</td></tr></table>	H	T	O	④	⑫		5	2	6	-	7	0				4	5	6	Tens	1 2	- 7		5
H	T	O																																					
6	9	4																																					
-	2	0																																					
6	7	4																																					
H	T	O																																					
④	⑫																																						
5	2	6																																					
-	7	0																																					
4	5	6																																					
Tens																																							
1 2																																							
- 7																																							
5																																							
Note: We only subtract tens.	Steps: (i) Subtract ones. (ii) We can not subtract 7 from 2 tens. So, we borrow 1 hundred leaving behind 4 hundreds. (iii) Now, subtract 7 tens from 12 tens to get 5.																																						

Subtraction of 2-digit number from 3-digit number.

Without regrouping

Example: Subtract 57 from 289

Solution:

$$\begin{array}{r} \text{H} \quad \text{T} \quad \text{O} \\ 2 \quad 8 \quad 9 \\ - \quad 5 \quad 7 \\ \hline 2 \quad 3 \quad 2 \end{array}$$

Note: We only subtract ones and tens separately.

With regrouping

Example: Subtract 64 from 532

Solution:

$$\begin{array}{r} \text{H} \quad \text{T} \quad \text{O} \\ \textcircled{4} \quad \textcircled{12} \quad \textcircled{12} \\ \cancel{5} \quad \cancel{3} \quad \cancel{2} \\ - \quad 6 \quad 4 \\ \hline 4 \quad 6 \quad 8 \end{array}$$

Steps:

- (i) We can not subtract 4 from 2. So, we borrow 1 ten leaving behind 2 tens.
- (ii) We subtract 4 from 12 to get 8.
- (iii) Borrow 1 hundred leaving behind 4 hundreds.
- (iv) Finally subtract 6 tens from 12 tens to get 6.

Subtraction of two 3-digit numbers.

Without regrouping

Example: Subtract 321 from 457

Solution:

$$\begin{array}{r}
 \text{H} \quad \text{T} \quad \text{O} \\
 4 \quad 5 \quad 7 \\
 - 3 \quad 2 \quad 1 \\
 \hline
 1 \quad 3 \quad 6
 \end{array}$$

Note: We only subtract digits of respective place values.

With regrouping

Example: Subtract 425 from 613

Solution:

$$\begin{array}{r}
 \text{H} \quad \text{T} \quad \text{O} \\
 \textcircled{5} \quad \textcircled{10} \quad \textcircled{13} \\
 6 \quad 1 \quad 3 \\
 - 4 \quad 2 \quad 5 \\
 \hline
 1 \quad 8 \quad 8
 \end{array}$$

Steps:

- (i) We borrow 1 ten leaving behind 0 ten to get 13 units.
- (ii) We subtract 5 units from 13 units to get 8 units.
- (iii) We borrow 1 hundred leaving behind 5 hundreds to get 10 tens.
- (iv) We subtract 2 tens from 10 tens to get 8 tens.
- (v) Finally subtract 4 hundreds from 5 hundreds to get 1 hundred.

Note: 10 tens = 1 hundred



Activity Subtract the following.

(a) 845

$- 9$

836

(b) 608

$- 15$

593

(c) 745

$- 30$

715

(d) 677

$- 288$

389

(e) 723

$- 6$

$\boxed{}$

(f) 510

$- 23$

$\boxed{}$

(g) 780

$- 123$

$\boxed{}$

(h) 645

$- 60$

$\boxed{}$

(i) 412

$- 123$

$\boxed{}$

(j) 678

$- 289$

$\boxed{}$

(k) 600

$- 78$

$\boxed{}$

(l) 962

$- 518$

$\boxed{}$

(m) 100

$- 8$

$\boxed{}$

(n) 500

$- 62$

$\boxed{}$

(o) 666

$- 277$

$\boxed{}$

(p) 852

$- 31$

$\boxed{}$

(q) 600

$- 60$

$\boxed{}$

(r) 223

$- 154$

$\boxed{}$

(s) 784

$- 10$

$\boxed{}$

(t) 602

$- 214$

$\boxed{}$

Exercise 3

1. Subtract mentally.

(i) $529 - 3 =$

(ii) $643 - 30 =$

(iii) $715 - 9 =$

(iv) $800 - 35 =$

(v) $692 - 215 =$

(vi) $750 - 60 =$

(vii) $752 - 121 =$

(viii) $845 - 369 =$

(ix) $509 - 36 =$

(x) $555 - 10 =$

2. Subtract the following.

(i) $\begin{array}{r} 235 \\ - \quad 2 \\ \hline \end{array}$

(ii) $\begin{array}{r} 506 \\ - \quad 39 \\ \hline \end{array}$

(iii) $\begin{array}{r} 785 \\ - 196 \\ \hline \end{array}$

(iv) $\begin{array}{r} 417 \\ - 250 \\ \hline \end{array}$

(v) $\begin{array}{r} 345 \\ - 60 \\ \hline \end{array}$

(vi) $\begin{array}{r} 715 \\ - \quad 8 \\ \hline \end{array}$

(vii) $\begin{array}{r} 405 \\ - 28 \\ \hline \end{array}$

(viii) $\begin{array}{r} 715 \\ - 426 \\ \hline \end{array}$

(ix) $\begin{array}{r} 600 \\ - 101 \\ \hline \end{array}$

(x) $\begin{array}{r} 800 \\ - 175 \\ \hline \end{array}$

(xi) $\begin{array}{r} 200 \\ - 99 \\ \hline \end{array}$

(xii) $\begin{array}{r} 754 \\ - 189 \\ \hline \end{array}$

Unit 2: Number Operations

Real life word problems with addition:

Example 1:

Ayesha purchased a book for Rs.560 and a note-book for Rs.280. How much did she spend in total?

Solution:

$$\begin{array}{rcl} \text{Cost of book} & = & 560 \text{ rupees} \\ \text{Cost of note-book} & = & + 280 \text{ rupees} \\ \hline \text{Total} & = & 840 \text{ rupees} \end{array}$$

Hence total amount she spent was 840 rupees.



Example 2:

Calculate the total weight of two packets of spices when their weights are 235g and 427g.

Solution:

$$\begin{array}{rcl} \text{Weight of 1}^{\text{st}} \text{ packet} & = & 235 \text{ g} \\ \text{Weight of 2}^{\text{nd}} \text{ packet} & = & + 427 \text{ g} \\ \hline \text{Total} & = & 662 \text{ g} \end{array}$$

Hence the total weight of two packets of spices was 662g.



Real life word problems with subtraction

Example 1:

There are 567 students in a school. If 229 are girls then how many are boys?

Solution:

$$\begin{array}{rcl} \text{Total students} & = & 567 \\ \text{Number of Girls} & = & - 229 \\ \hline \text{Number of Boys} & = & 338 \end{array}$$

Hence there are 338 boys in the school.



Example 2:

A factory produces 562 bikes in a month. If 491 are sold, how many bikes are left?

Solution:

$$\begin{array}{r}
 \text{Total bikes} = 562 \\
 \text{Bikes sold} = - 491 \\
 \hline
 \text{Bikes left} = 71
 \end{array}$$

Hence 71 bikes are left.

**Exercise 4**

1. Bisma has two number cards having numbers 278 and 356. What is their sum?
2. Abdul Rafay purchased a pen for Rs. 135 rupees and a bag for Rs. 675 rupees. How much did he spend?
3. 236 people visited zoo on Saturday and 417 people visited on Sunday. How many people did visit the zoo altogether.
4. There were 157 balloons in the birthday party of Taha. If 129 balloons were burst by the children then how many are left?
5. Sum of two numbers is 786. If one is 345 then what is the other number?
6. Bushra and Aafia arranged a party for children. Total expenses were 950 rupees. How much did Bushra spend if Aafia spent 460 rupees?
7. A shopkeeper sold 135 liters of milk on first day and 204 liters on second day. How much milk did he sold in two days?
8. Tanveer had 875 rupees. He spent 350 rupees on ice-cream. How much amount is left with him.

Estimating the answer to an addition and subtraction

Estimation means to get close enough to exact answer

We can estimate sum or difference by rounding to the desired digit or by front-end estimation.

Example 1:

Estimate the sum: $241 + 36$ by rounding to the nearest tens.

Solution:

Rounding to nearest tens

$241 \rightarrow 240$ $\downarrow$ (round down)

$36 \rightarrow 40$ $\uparrow$ (round up)

Addition

$$\begin{array}{r} 240 \\ + 40 \\ \hline 280 \end{array}$$

So, the required estimation is 280.

Example 2:

Estimate the difference $675 - 34$ by rounding to the nearest tens.

Solution:

Rounding to nearest tens

$675 \rightarrow 680$ $\uparrow$ (round up)

$34 \rightarrow 30$ $\downarrow$ (round down)

Subtraction

$$\begin{array}{r} 680 \\ - 30 \\ \hline 650 \end{array}$$

So, the required estimation is 650.

(By front-end estimation)

Front-end estimation means highest place value estimation.

Example 3:

Estimate sum and difference of 583 and 216 by front-end estimation.

Solution:

Rounding to front-end

$$583 \rightarrow 600$$

 $\boxed{\uparrow}$ (round up)

$$216 \rightarrow 200$$

 $\boxed{\downarrow}$ (round down)

$$\begin{array}{r} \text{Addition:} \quad 6 \ 0 \ 0 \\ + \quad 2 \ 0 \ 0 \\ \hline 8 \ 0 \ 0 \end{array}$$

Estimated sum = 800

$$\begin{array}{r} \text{Subtraction:} \quad 6 \ 0 \ 0 \\ - \quad 2 \ 0 \ 0 \\ \hline 4 \ 0 \ 0 \end{array}$$

Estimated difference = 400

Exercise 5

1. Estimate the following by rounding nearest to tens.

46 + 58 = _____	46 - 34 = _____
462 + 34 = _____	372 - 18 = _____
421 + 567 = _____	884 - 625 = _____
588 + 461 = _____	965 - 486 = _____

2. Estimate the following sum and difference by front-end estimation.

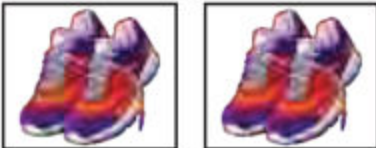
49 + 38 = _____	56 - 31 = _____
32 + 75 = _____	55 - 11 = _____
615 + 456 = _____	708 - 178 = _____
776 + 220 = _____	333 - 266 = _____

Unit 2: Number Operations

Even number:

The number that can produce pairs of objects with no remainder.

For example:

Even number	Objects
4	
6	

Odd number:

The number that can not produce pairs of objects with no remainder.

For example:

Odd number	Objects
5	
7	

Example 1:

Write odd numbers between 20 and 30.

Solution:

20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30

Odd numbers are: 21, 23, 25, 27 and 29

Example 2:

Write even numbers from 50 to 60.

Solution:

50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60

Even numbers are: 50, 52, 54, 56, 58 and 60

Even and odd numbers up to 100.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

In the above table numbers of blue colour are odd and the numbers in orange colour are even.

Notice that:

Odd number	has	1, 3, 5, 7 or 9	at unit place
------------	-----	-----------------	---------------

Even number	has	0, 2, 4, 6 or 8	at unit place
-------------	-----	-----------------	---------------



Activity

Write even or odd for the following

56	Even
39	Odd
20	
99	
64	
71	
82	
19	
24	

Exercise 6

1. Separate even and odd numbers.



Even numbers: 8, _____

Odd numbers: 5, _____

2. Encircle even numbers from the following.

31, 42, 17, 27, 64, 89, 96, 18, 13, 100

3. Encircle odd numbers from the following.

16, 91, 83, 78, 22, 41, 58, 69, 76

4. Write odd numbers between 30 and 40.

5. Write even numbers between 51 and 61.

Counting in 3s.



Activity

Count and write in 3s.



**Activity**

Count in 3s and write the missing numbers.

15	18		24			33
9	12			21	24	

Counting in 4s.**Activity**

Count and write in 4s.

4	8	12	16	20	24
100	104	108	112	116	120

**Activity**

Count in 4s and write the missing numbers.

16	20		28		
28		36		44	

Counting in 5s.**Activity**

Count and write in 5s.

5	10	15	20	25	30	35
60	65	70	75	80	85	90



Activity Count in 5s and write the missing numbers.

40	45			60		70
105		115	120		130	

Counting in 10s.



Activity Count and write in 10s.

10	20	30	40	50	60	70
80	90	100	110	120	130	140



Activity Count in 10s and write the missing numbers.

50	60			90		
110			140			170

Counting in 100s.



Activity Count and write in 100s.

100	200	300	400	500	600
105	205	305	405	505	605



Activity

Count in 100s and write the missing numbers.

500				900
200			400	

Exercise 7

1. Count in 3s and write the missing numbers.

15		21			30
30		36		42	

2. Count in 4s and write the missing numbers.

8		16		24	
0		48			60

3. Count in 5s and write the missing numbers.

15		25		35	
110		120		130	

4. Count in 10s and write the missing numbers.

40		60		80	
220		240			270

Unit 2: Number Operations

5. Count in 100s and write the missing numbers.

300		500			800
220		420		620	



Activity

Count in 3s (threes) and write.

Objects	Counting
	1 three is 3
 	2 threes are 6
  	3 threes are 9

Similarly, 4 threes are 12

5 threes are 15

6 threes are 18

7 threes are 21

8 threes are 24

9 threes are 27

10 threes are 30

Multiplication table of 3

1	three is	3
2	threes are	6
3	threes are	9
4	threes are	12
5	threes are	15
6	threes are	18
7	threes are	21
8	threes are	24
9	threes are	27
10	threes are	30



Activity Count in 4s (fours) and write.

Objects	Counting
	1 four is 4
	2 fours are 8
	3 fours are 12

Similarly,

4	Fours are	16
5	Fours are	20
6	Fours are	24
7	Fours are	28
8	Fours are	32
9	Fours are	36
10	Fours are	40

Multiplication table of 4

1	four is	4
2	fours are	8
3	fours are	12
4	fours are	16
5	fours are	20
6	fours are	24
7	fours are	28
8	fours are	32
9	fours are	36
10	fours are	40

Exercise 8

1. Match the following.

3 threes	15
5 threes	30
6 threes	9
9 threes	27
10 threes	18

2. Match the following.

3 fours	32
5 fours	36
7 fours	12
8 fours	20
9 fours	28

3. Complete the following.

3 fours are 12

6 fours are

5 threes are

9 threes are

5 fours are

8 fours are

2 threes are

Multiplication as repeated addition

Look at these cars.



How many wheels are in total?

We just add 4 repeatedly

$$4 + 4 + 4 = 12$$

Or

three times 4 is 12

We write it as

$$3 \times 4 = 12$$

It is read as "3 multiply 4 is equal to 12"

Note: " $\times$ " is the sign of multiplication.



Activity Count the objects and write.

Objects	Counting
	$2 + 2 + 2 + 2 = 8$ or $4 \text{ times } 2 = 8$ or $4 \times 2 = 8$
	$3 + 3 + 3 + 3 + 3 = 15$ or $5 \text{ times } 3 = 15$ or $5 \times 3 = 15$
	$4 + 4 + 4 + 4 + 4 = 20$ or $5 \text{ times } 4 = 20$ or $5 \times 4 = 20$



Activity Match the following:

$5 + 5 + 5 + 5$	5×6
$2 + 2 + 2 + 2 + 2$	8×4
$3 + 3 + 3 + 3 + 3 + 3$	4×5
$4 + 4 + 4 + 4 + 4 + 4 + 4 + 4$	5×2
$6 + 6 + 6 + 6 + 6$	6×3

Multiplication Tables

Table of 2

Repeated addition of 2	Table of 2	Table of 2
2	1 two is 2	$1 \times 2 = 2$
2 + 2	2 twos are 4	$2 \times 2 = 4$
2 + 2 + 2	3 twos are 6	$3 \times 2 = 6$
2 + 2 + 2 + 2	4 twos are 8	$4 \times 2 = 8$
2 + 2 + 2 + 2 + 2	5 twos are 10	$5 \times 2 = 10$
2 + 2 + 2 + 2 + 2 + 2	6 twos are 12	$6 \times 2 = 12$
2 + 2 + 2 + 2 + 2 + 2 + 2	7 twos are 14	$7 \times 2 = 14$
2 + 2 + 2 + 2 + 2 + 2 + 2 + 2	8 twos are 16	$8 \times 2 = 16$
2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2	9 twos are 18	$9 \times 2 = 18$
2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2	10 twos are 20	$10 \times 2 = 20$

Table of 3

Repeated addition of 3	Table of 3	Table of 3
3	1 three is 3	$1 \times 3 = 3$
3 + 3	2 threes are 6	$2 \times 3 = 6$
3 + 3 + 3	3 threes are 9	$3 \times 3 = 9$
3 + 3 + 3 + 3	4 threes are 12	$4 \times 3 = 12$
3 + 3 + 3 + 3 + 3	5 threes are 15	$5 \times 3 = 15$
3 + 3 + 3 + 3 + 3 + 3	6 threes are 18	$6 \times 3 = 18$
3 + 3 + 3 + 3 + 3 + 3 + 3	7 threes are 21	$7 \times 3 = 21$
3 + 3 + 3 + 3 + 3 + 3 + 3 + 3	8 threes are 24	$8 \times 3 = 24$
3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3	9 threes are 27	$9 \times 3 = 27$
3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3	10 threes are 30	$10 \times 3 = 30$

Table of 4

Repeated addition of 4	Table of 4	Table of 4
4	1 four is 4	$1 \times 4 = 4$
4 + 4	2 fours are 8	$2 \times 4 = 8$
4 + 4 + 4	3 fours are 12	$3 \times 4 = 12$
4 + 4 + 4 + 4	4 fours are 16	$4 \times 4 = 16$
4 + 4 + 4 + 4 + 4	5 fours are 20	$5 \times 4 = 20$
4 + 4 + 4 + 4 + 4 + 4	6 fours are 24	$6 \times 4 = 24$
4 + 4 + 4 + 4 + 4 + 4 + 4	7 fours are 28	$7 \times 4 = 28$
4 + 4 + 4 + 4 + 4 + 4 + 4 + 4	8 fours are 32	$8 \times 4 = 32$
4 + 4 + 4 + 4 + 4 + 4 + 4 + 4 + 4	9 fours are 36	$9 \times 4 = 36$
4 + 4 + 4 + 4 + 4 + 4 + 4 + 4 + 4 + 4	10 fours are 40	$10 \times 4 = 40$

Table of 5

Repeated addition of 5	Table of 5	Table of 5
5	1 five is 5	$1 \times 5 = 5$
5 + 5	2 fives are 10	$2 \times 5 = 10$
5 + 5 + 5	3 fives are 15	$3 \times 5 = 15$
5 + 5 + 5 + 5	4 fives are 20	$4 \times 5 = 20$
5 + 5 + 5 + 5 + 5	5 fives are 25	$5 \times 5 = 25$
5 + 5 + 5 + 5 + 5 + 5	6 fives are 30	$6 \times 5 = 30$
5 + 5 + 5 + 5 + 5 + 5 + 5	7 fives are 35	$7 \times 5 = 35$
5 + 5 + 5 + 5 + 5 + 5 + 5 + 5	8 fives are 40	$8 \times 5 = 40$
5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5	9 fives are 45	$9 \times 5 = 45$
5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5	10 fives are 50	$10 \times 5 = 50$

Table of 10

Repeated addition of 10	Table of 10	Table of 10
10	1 ten is 10	$1 \times 10 = 10$
$10 + 10$	2 tens are 20	$2 \times 10 = 20$
$10 + 10 + 10$	3 tens are 30	$3 \times 10 = 30$
$10 + 10 + 10 + 10$	4 tens are 40	$4 \times 10 = 40$
$10 + 10 + 10 + 10 + 10$	5 tens are 50	$5 \times 10 = 50$
$10 + 10 + 10 + 10 + 10 + 10$	6 tens are 60	$6 \times 10 = 60$
$10 + 10 + 10 + 10 + 10 + 10 + 10$	7 tens are 70	$7 \times 10 = 70$
$10 + 10 + 10 + 10 + 10 + 10 + 10 + 10$	8 tens are 80	$8 \times 10 = 80$
$10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10$	9 tens are 90	$9 \times 10 = 90$
$10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10$	10 tens are 100	$10 \times 10 = 100$

Exercise 9

1. Complete the following.

$6 + 6 + 6 + 6$

4 times 6

4×6

$2 + 2 + 2 + 2 + 2$

5 times 2

$3 + 3 + 3 + 3 + 3 + 3$

6×3

3 times 7

3×7

$5 + 5 + 5 + 5$

2. Fill in the boxes.

$$2 \times 5 = 10$$

$$3 \times 2 = \boxed{}$$

$$6 \times 4 = \boxed{}$$

$$7 \times \boxed{} = 70$$

$$8 \times \boxed{} = 32$$

$$\boxed{} \times 3 = 27$$

$$\boxed{} \times 5 = 30$$

$$6 \times 3 = \boxed{}$$

$$7 \times 3 = \boxed{}$$

$$9 \times 4 = \boxed{}$$

$$3 \times 5 = \boxed{}$$

$$6 \times 10 = \boxed{}$$

$$7 \times \boxed{} = 14$$

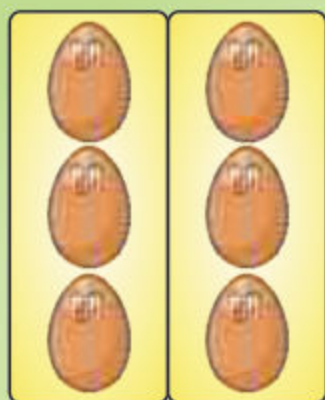
$$\boxed{} \times 4 = 40$$

Multiplication statements make counting easy.



Activity

Count and write the number of objects.



Two times three is 6

$$2 \times 3 = 6$$



Three times four is 12

$$3 \times 4 = 12$$

Multiplication statements


Activity

Using given figures, match the following.

Figure 1	Figure 2	Figure 3

3×5	Figure 1
4×3	Figure 2
5×4	Figure 3

An arrow points from 3×5 to Figure 2.

Exercise 10

Count and write multiplication statements for the following.

Objects	Multiplication statements
1.	Four times three is 12 or $4 \times 3 = 12$

In multiplication, the result of any two numbers is always same in any order.

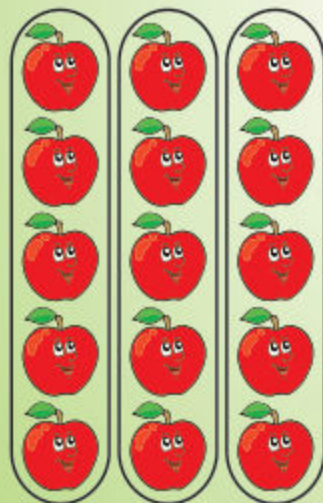


Activity Count the objects and write.



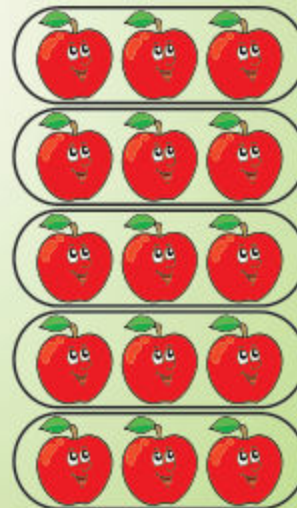
We can count in two ways

First way



$$3 \times 5 = 15$$

Second way



$$5 \times 3 = 15$$

i.e. $3 \times 5 = 15 = 5 \times 3$

Similarly, $2 \times 3 = 6 = 3 \times 2$

Exercise 11

1. Complete the following.

(i) $5 \times 3 = \boxed{3} \times 5$

(ii) $10 \times 4 = 4 \times \boxed{}$

(iii) $3 \times 6 = \boxed{} \times 3$

(iv) $\boxed{} \times 4 = 4 \times 5$

(v) $7 \times \boxed{} = 2 \times 7$

2. Match the following

3×4

6×4

2×10

4×10

4×6

7×5

5×7

4×3

10×4

10×2

We use multiplication tables to multiply one number by another.

Example 1:

Multiply 5 by 2

Solution:

Here, we read table of 2 up to 5 and get 10.

So,

$$5 \times 2 = 10 \quad \text{or}$$

	5
$\times$	2
1	0

Table of 2

$1 \times 2 = 2$
$2 \times 2 = 4$
$3 \times 2 = 6$
$4 \times 2 = 8$
$5 \times 2 = 10$
$6 \times 2 = 12$
$7 \times 2 = 14$
$8 \times 2 = 16$
$9 \times 2 = 18$
$10 \times 2 = 20$

Example 2:

Multiply 6 by 3

Solution:

Here, we read table of 3 up to 6 and get 18.

So,

$$6 \times 3 = 18 \quad \text{or}$$

	6
$\times$	3
1	8

Table of 3

$1 \times 3 = 3$
$2 \times 3 = 6$
$3 \times 3 = 9$
$4 \times 3 = 12$
$5 \times 3 = 15$
$6 \times 3 = 18$
$7 \times 3 = 21$
$8 \times 3 = 24$
$9 \times 3 = 27$
$10 \times 3 = 30$



Activity Solve the following

5
$\times$ 4
2 0

8
$\times$ 5

9
$\times$ 2

7
$\times$ 3

Exercise 12

1. Multiply

(i) 6 by 3.

(iii) 4 by 5.

(v) 9 by 2.

(ii) 7 by 2.

(iv) 5 by 5.

(vi) 8 by 4.

2. Fill in the blanks.

(i) $5 \times 3 =$

(ii) $6 \times 4 =$

(iii) $9 \times 5 =$

(iv) $7 \times 2 =$

(v) $3 \times 4 =$

(vi) $9 \times 3 =$

3. Solve

$$\begin{array}{r} 4 \\ \times 2 \\ \hline \end{array}$$

$$\begin{array}{r} 5 \\ \times 3 \\ \hline \end{array}$$

$$\begin{array}{r} 9 \\ \times 4 \\ \hline \end{array}$$

$$\begin{array}{r} 8 \\ \times 5 \\ \hline \end{array}$$

$$\begin{array}{r} 7 \\ \times 4 \\ \hline \end{array}$$

$$\begin{array}{r} 8 \\ \times 2 \\ \hline \end{array}$$

$$\begin{array}{r} 4 \\ \times 3 \\ \hline \end{array}$$

$$\begin{array}{r} 9 \\ \times 2 \\ \hline \end{array}$$

$$\begin{array}{r} 5 \\ \times 5 \\ \hline \end{array}$$

$$\begin{array}{r} 7 \\ \times 5 \\ \hline \end{array}$$

$$\begin{array}{r} 3 \\ \times 3 \\ \hline \end{array}$$

$$\begin{array}{r} 4 \\ \times 4 \\ \hline \end{array}$$

Multiplication using multiplication grid

In multiplication grid, we follow the following steps:

1. Partition a 2-digit number into tens and ones.
2. Multiply numbers in grid.
3. Add the results.

Example 1:

Multiply 25 with 3

Solution: Here, $25 = 20 + 5$

Multiplication Grid

×	20	5
3	60	15

$$\begin{aligned} \text{So, } 25 \times 3 &= 60 + 15 \\ &= 75 \end{aligned}$$

$$\begin{array}{r} 60 \\ + 15 \\ \hline 75 \end{array}$$

Explanation:

- First, we partition 25 into 20 and 5.
- We write 20 and 5 in top row and 3 in the first column.
- We multiply 3 by 20 to get 60 and 3 by 5 to get 15.
- Finally, we add 60 and 15.

Example 2:

Multiply 34 by 2

Solution: Here, $34 = 30 + 4$

Multiplication Grid

×	30	4
2	60	8

$$\begin{aligned} \text{So, } 34 \times 2 &= 60 + 8 \\ &= 68 \end{aligned}$$

$$\begin{array}{r} 60 \\ + 8 \\ \hline 68 \end{array}$$

Unit 2: Number Operations

Multiply of a number with 0 and 1.

Three children are asking bananas from their mother.



Ayesha has 0 banana
Taha has 0 banana
Aafia has 0 banana
Total = $0 + 0 + 0 = 0$
or $3 \times 0 = 0$

i.e. **Zero multiplied by any number is zero.**

Now, mother has given them 1 banana to each



Ayesha has 1 banana
Taha has 1 banana
Aafia has 1 banana
Total = $1 + 1 + 1 = 3$
or $3 \times 1 = 3$

i.e. **One multiplied by any number is that number**

Example:

i) Multiply 6 by 0

$$6 \times 0 = \boxed{0}$$

ii) Multiply 5 by 1

$$5 \times 1 = \boxed{5}$$

Exercise 13

1. Multiply the following using multiplication grid.

(i) 17 by 2

(ii) 32 by 3

(iii) 26 by 4

(iv) 12 by 5

(v) 43 by 2

(vi) 24 by 3

(vii) 27 by 2

(viii) 35 by 2

2. Multiply

(i) 6 by 0

(ii) 7 by 1

(iii) 8 by 0

(iv) 4 by 1

(v) 5 by 0

(vi) 8 by 1

3. Fill in the blanks.

(i) $5 \times 0 = \boxed{}$

(ii) $7 \times 0 = \boxed{}$

(iii) $0 \times 4 = \boxed{}$

(iv) $0 \times 2 = \boxed{}$

(v) $2 \times 1 = \boxed{}$

(vi) $9 \times 1 = \boxed{}$

(vii) $1 \times 5 = \boxed{}$

(viii) $1 \times 3 = \boxed{}$

Real life problem

Example 1:

The price of one pencil is Rs. 10. Find the price of such 6 pencils.



$$\begin{aligned} \text{Price of one pencil} &= 10 \text{ rupees} \\ \text{So, Price of six pencils} &= 6 \times 10 \\ &= 60 \end{aligned}$$

1	0
$\times$	6
6	0

So, the price of 6 pencils is Rs. 60.

Example 2:

There are 4 wheels in a van. How many wheels are there in 7 vans.

$$\begin{aligned} \text{Number of wheels in a van} &= 4 \\ \text{So, Number of wheels in 7 vans} &= 7 \times 4 \\ &= 28 \end{aligned}$$

	7
$\times$	4
2	8

So, there are 28 wheels in 7 cars.

Exercise 14

1. The price of a chocolate is Rs. 5. What is the price of such 9 chocolate?
2. There are 4 fans in a room. How many fans are there in such 8 rooms?
3. The price of an egg is Rs.22. What is the price of 6 eggs?
4. Abdul Rafay spends Rs.35 in a day. How much does he spend in 7 days?
5. There are 4 persons in a car. How many persons are in 5 cars?
6. A shirt has 8 buttons. How many buttons are there in 5 shirts?
7. A dog has 4 legs. How many legs do 7 dogs have?
8. The price of a sharpener is Rs.8. What is the price of such 9 sharpeners?

Division



Activity

Distribute 12 mangoes equally among 3 children, Bushra, Tanveer and Ayesha.



Sonia



Tanveer



Ayesha

Each child gets 4 mangoes. We have infact shared 12 mangoes among 3 children.

Division is equal sharing number of objects

As multiplication is repeated addition. Similarly division is repeated subtraction. Let us see in the following example.

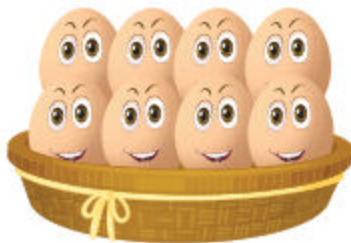
Example:

Divide 10 eggs equally in 2 baskets by repeated subtraction.



Solution:

Let us subtract 2 eggs again and again.



$$10 - 2 = 8$$

Unit 2: Number Operations



$$8 - 2 = 6$$



$$6 - 2 = 4$$



$$4 - 2 = 2$$



$$2 - 2 = 0$$

Let us count how many times we have subtracted 2.
We have subtracted 5 times.
So, 10 divided by 2 is 5
i.e. $10 \div 2 = 5$
Note: Symbol of division is " $\div$ "

Exercise 15

1. Divide 12 balls equally in 3 ball stands by repeated subtraction.



$$12 - 3 = 9$$



$$9 - 3 = 6$$



$$6 - 3 = 3$$



$$3 - 3 = 0$$

So, $12 \div 3 =$

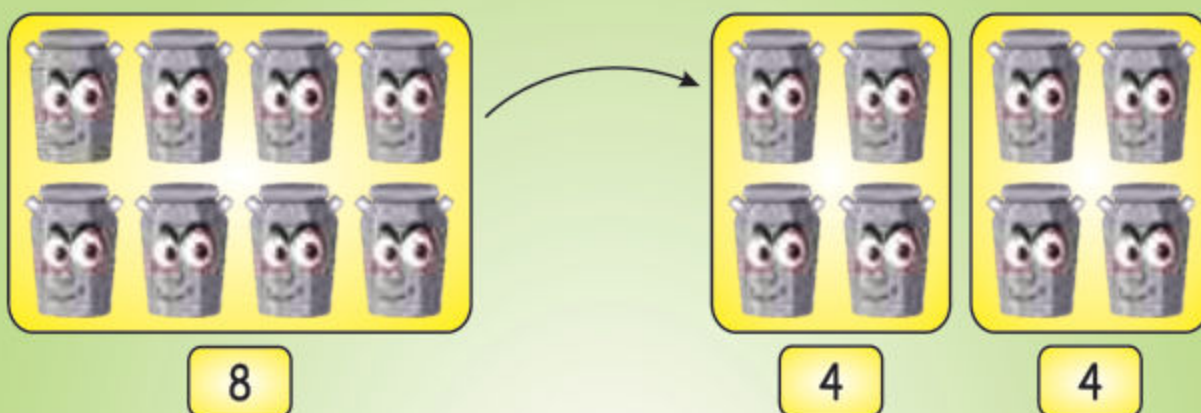
Division statements



Activity

Write division statements for the following groupings.

1.



Here 8 objects distributed equally in 2 groups of 4 objects.

i.e. 8 divided by 2 is 4

or $8 \div 2 = 4$

2.



Here 6 objects distributed equally in 2 groups of 3 objects.

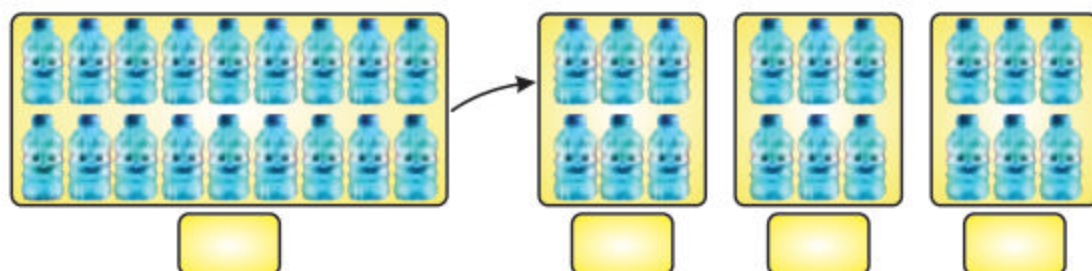
i.e. 6 divided by 2 is 3

or $6 \div 2 = 3$

Exercise 16

Write division statements for the following groupings

1.

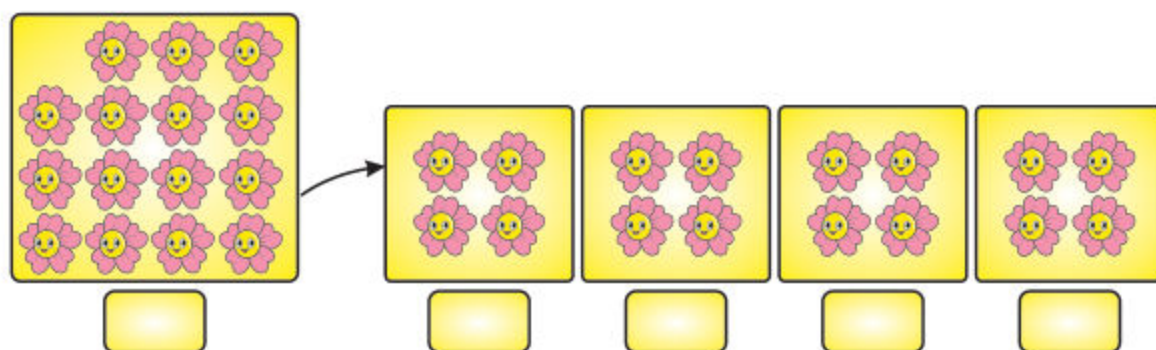


18 divided by 3 is 6

i.e.

$$18 \div 3 = 6$$

2.

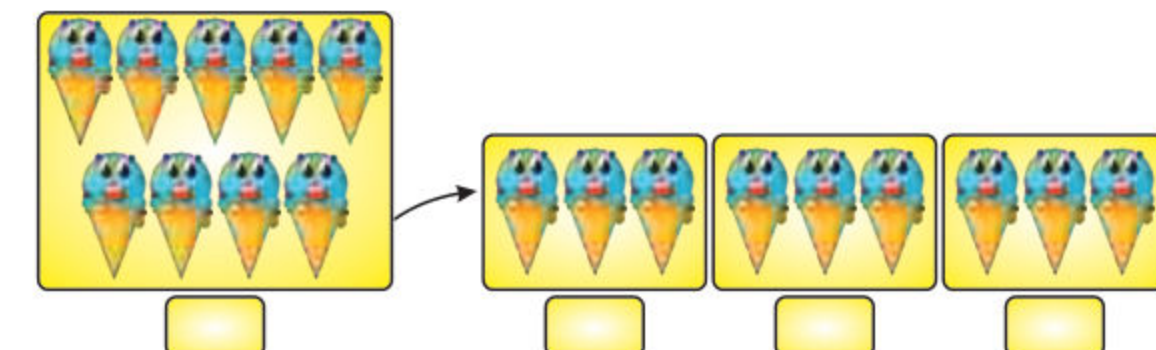


16 divided by 4 is

i.e.

$$\square \div \square = \square$$

3.



9 divided by 3 is

i.e.

$$\square \div \square = \square$$

We know that multiplication can be done in any order for example:

$$4 \times 2 = 8 = 2 \times 4$$

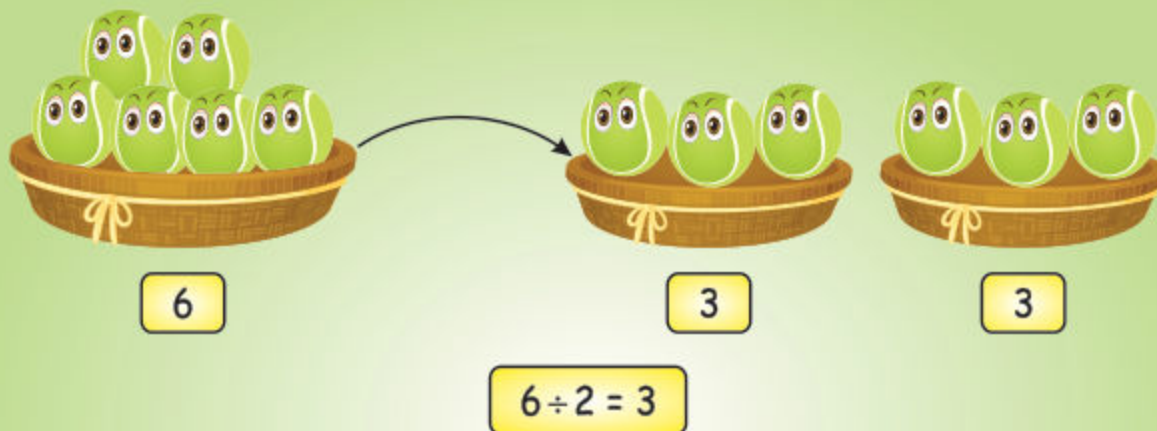
But division can not be done in any order.

Let us see by the following activity.



Activity

Divide 6 balls in two baskets .



If we try to divide 2 balls in six baskets. Let see what happen



We can not divide 2 balls in 6 baskets because balls are less and baskets are more.

So,

Hence division can not be done in any order.

Exercise 17

1. Can we divide 8 books equally among 12 students?

Yes

or

No



2. Can we divide 5 balls equally among 5 students?

Yes

or

No

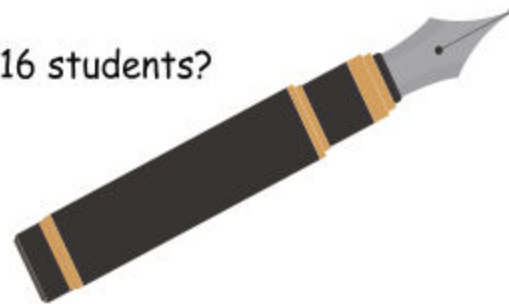


3. Can we divide 8 pens equally among 16 students?

Yes

or

No

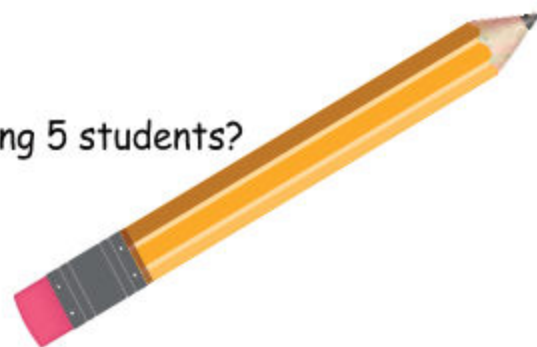


4. Can we divide 15 pencils equally among 5 students?

Yes

or

No



5. Can we divide 12 chocolates equally among 20 students?

Yes

or

No



Unit 2: Number Operations

Division of 1-digit and 2-digit numbers by 1-digit number (without remainder)

We have already studied that division means the equal sharing of objects without remainder.

$$6 \div 2 = 3 \text{ (without remainder)}$$

We can also write it as under:

$$\begin{array}{r} 3 \\ 2 \overline{) 6} \\ \underline{- 6} \\ 0 \end{array}$$

Labels: Quotient (3), Divisor (2), Dividend (6), Remainder (0)

In order to divide 6 by 2 we read table of 2 till 6 comes.

i.e. $2 \times 3 = 6$

So, $6 \div 2 = 3$

Note: Division is the reverse process of multiplication.

Example Divide:

1. $8 \div 2 = 4$ because $2 \times 4 = 8$

2. $9 \div 3 = 3$ because $3 \times 3 = 9$

3. $15 \div 5 = 3$ because $5 \times 3 = 15$

4. $20 \div 4 = 5$ because $4 \times 5 = 20$

Example Divide:

1.
$$\begin{array}{r} 2 \\ 3 \overline{) 6} \\ \underline{- 6} \\ 0 \end{array}$$

2.
$$\begin{array}{r} 2 \\ 2 \overline{) 4} \\ \underline{- 4} \\ 0 \end{array}$$

3.
$$\begin{array}{r} 7 \\ 3 \overline{) 21} \\ \underline{- 21} \\ 0 \end{array}$$

4.
$$\begin{array}{r} 6 \\ 4 \overline{) 24} \\ \underline{- 24} \\ 0 \end{array}$$

Exercise 18

1. Divide:

i) $6 \div 3 = 2$ because $2 \times 3 = 6$

ii) $8 \div 4 =$ because

iii) $24 \div 6 =$ because

iv) $28 \div 4 =$ because

v) $36 \div 6 =$ because

2. Divide:

1.

$$\begin{array}{r} 2 \overline{) 8} \\ \underline{} \\ \end{array}$$

2.

$$\begin{array}{r} 2 \overline{) 8} \\ \underline{} \\ \end{array}$$

3.

$$\begin{array}{r} 4 \overline{) 24} \\ \underline{} \\ \end{array}$$

4.

$$\begin{array}{r} 10 \overline{) 20} \\ \underline{} \\ \end{array}$$

All real life word problems

Example 1:

An student gets 15 marks in each of 5 subjects. What are his total marks?

Solution:

Number of subjects = 5

Marks in each subject = 15

$$\begin{aligned} \text{Required number of marks} &= 15 \times 5 \\ &= 75 \end{aligned}$$

So, his total marks are 75 in 5 subjects.

Working

$\times$	10	5
5	50	25

$$\begin{array}{r} 50 \\ + 25 \\ \hline 75 \end{array}$$

Example 2:

50 chocolates are distributed equally among 10 students. How many chocolates does each child get?

Solution:

Number of chocolates = 50

Number of students = 10

Required number of chocolates = $50 \div 10$
= 5

So, each child gets 5 chocolates.



Exercise 19

1. 18 apples are equally distributed to 3 students. How many apples does each student get?
2. There are 12 trees. On each tree, 4 birds are sitting. How many birds are sitting on all the trees?
3. 16 books are distributed equally among 4 children. How many books does each child get?
4. If price of a pen is 25 rupees. What will be the price of 12 such pens?
5. If shazia gets 20 rupees as her pocket money daily. How much will she get after 15 days?

Unit 3

MONEY

Student Learning Outcomes:

At the end of the chapters, Students will be able to:

- ✓ Identify international currency and denominations (for instance, dollar, yuan and riyal).
- ✓ Solve money-related problems involving the addition and subtraction of Pakistani money and a few selected international currency notes (for instance; dollar, yuan, and riyal).
- ✓ Solve real life word problems (including Pakistani currency) involving addition, subtraction, multiplication and division.

